

Project 4: Autonomous Mobile Pick-and-Place Robot

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1. Project Overview and Introduction

1a. Project objective:

Design an autonomous wheeled robot that picks up three colored cubes from Spot A and places them into the corresponding cups at Spot B in the specified color and order sequence.

1b. Problem statement:

The system must autonomously navigate between two points and perform pick-and-place operations without human intervention after the start command.

1c. Motivation and relevance to mechatronics:

This project is relevant to mechatronics because it integrates mechanical design, sensors, actuators, power electronics, and control logic into one autonomous system. That matches the course requirement that each project include core subsystems for mechanical design, sensing, actuation, and control/software.

1d. Summary of system functionality:

The robot must travel along a path, identify or otherwise handle three colored cubes, pick them up, transport them, and place them into matching cups at the destination. It must do this using onboard power only and without cameras or vision systems.

1e. Key results:

The robot functions per the project parameters with little to no assistance with picking up the colored cubes, which can be placed in any color order and still dropped

off into their respective cups. All of this is done autonomously using its onboard power supplies, which are discussed further in this document.

2. System Architecture and Design

2a. Overall system block diagram:

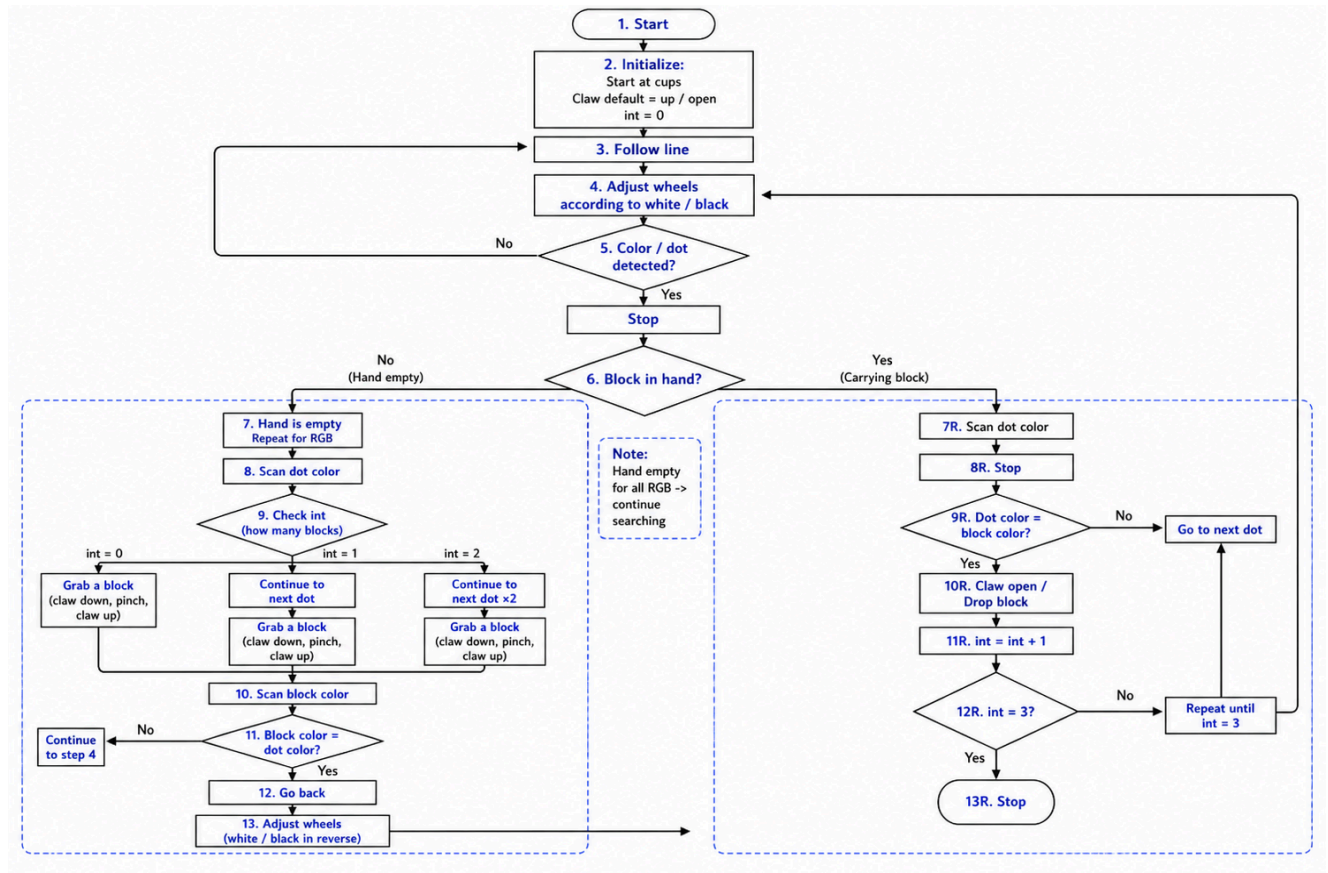


Figure (1): System Block Diagram

2b. Mechanical subsystem description:

Four wheeled (two powered) mobile base for locomotion and a grasping claw mechanism for cube pickup and placement that moves up and down via ball-screw system.

2c. Sensing subsystem description:

The robot must use some non-vision sensing approach because cameras and vision systems are not allowed, so two different sensors were implemented. The first sensor is a TCS3200 color sensor (5V and 3 wavelength channels), which is used to sense the block color in the claw attachment. Whereas the second sensor is an AS7262

spectral sensor (3.3V and 6 wavelength channels), which is used to read the following line and the ground colors that signal changes in the robot's behavior.

2d. Actuation subsystem description:

The system requires actuators for robot movement, ball-screw actuation, and grasping mechanism. For the movement, two wheels were each hooked up to their own VEX EDR 393 geared DC motor, where they are both driven by a single 105090004 DC motor driver and powered by a 7.4V rechargeable battery. Similarly, the ball-screw actuation is controlled by an SY42STH38-1684A stepper motor, which is driven by an A4988 stepper motor driver and also powered by a 7.4V rechargeable battery. Lastly, the grasping mechanism is a 20Kg DS3218 servo motor driven by our Arduino Uno and powered by a quad AA battery module generating 5.8V. All of this can be visualized in the following *Table (1)* for simplicity:

Table (1): Table of actuators with their respective drivers and power supplies

Component	Usage	Driver	Voltage Required	Power Source
Arduino	Brain	N/A	5.0V	9V Battery
20kg Servo Motor	Gripping Claw	Arduino	6.8V	X4 AA Battery Pack
Stepper Motor	Ball-Screw Mechanism	A4988	7.4V	7.4V LIPO Battery
X2 Geared DC Motors	Left/Right Wheels	105090004	7.4V	7.4V LIPO Battery

2e. Control subsystem description:

The control subsystem manages the robot's movement, sensing, decision-making, and claw operation. The robot starts at the cups with the claw in the up/open position and follows the line by adjusting the drive wheels based on black/white line sensor readings. When a color dot is detected, the robot stops and checks whether it is holding a block. If the claw is empty, the controller scans the dot color, checks the current int value, and uses it to decide where to pick up the next block. Depending on whether $int = 0$, $int = 1$, or $int = 2$, the robot either grabs a block immediately, at the next dot, or after two dots. The claw then moves down, pinches the block, and lifts back up. The block color is scanned and compared to the dot color. If the colors match, the robot returns along the line in reverse and searches for the matching drop-off dot. If the colors

do not match, it resumes line-following and continues searching. When carrying a block, the robot scans each dot, stops, and compares the dot color to the block color. If the colors match, the claw opens and drops the block. The controller then increments int by one and repeats the process until $\text{int} = 3$, at which point the robot stops.

2f. Design constraints and requirements:

The design constraints are as follows: 100 dollar budget total, fully autonomous after start, no cameras or vision systems, onboard power only, must pick and place three colored cubes, must place cubes into corresponding cups at location B in the specified color/order sequence, needs to repeat this process three times for each cube reliably with no assistance, entire unit must weigh less than the maximum output torque of the wheel motors.

3. Design Decisions and Engineering Justification

3a. Decision matrix comparing alternative mechanisms, sensors, and actuators:

For our arm designs, we narrowed the options down to three: an articulated arm, a ball screw arm, and an extending arm. The articulated works by using servos and linkages to transfer the motion to extend the arm. The ball screw used a stepper motor and a ball screw to raise and lower the claw. Finally, the extending arm uses a linear actuator to extend the arm forward and grab the block.

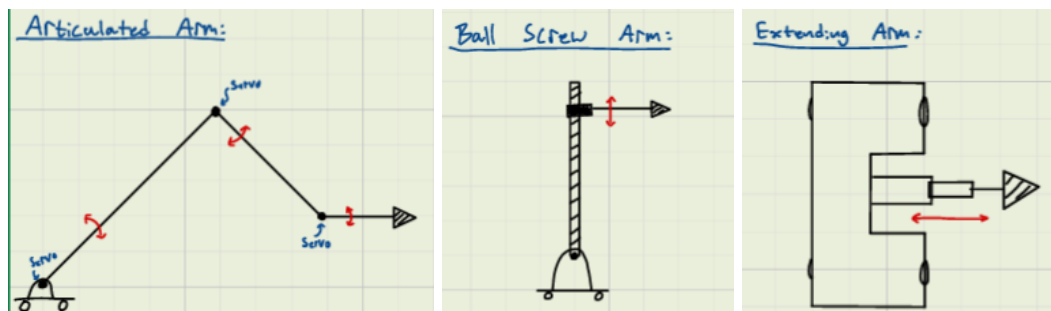


Figure (2): Potential Arm Designs

Table (2): Potential Arm Design Decision Matrix

Arm	Price	Ease	Redundancy	Reliability	Total
Articulated	4	2	1	3	10
Ball-Screw	3	4	4	4	15
Extending	2	3	4	4	12

Decision: Ball Screw

For our hand design options, we pick the top three to decide from: a vice grip clamp, a servo clamp, and a linkage articulated clamp. The vice grip clamp uses a stepper to drive a screw design to clamp onto a block. The servo clamp uses two servos to drive two linkages. The linkage articulated clamp uses a servo connected to a geared parallel linkage to extend the clamp forward and grab a block.

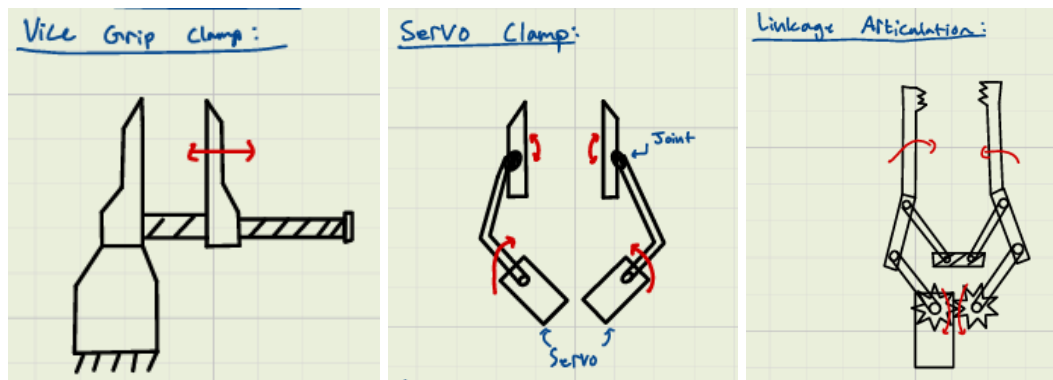


Figure (3): Potential Claw Designs

Table (3): Potential Claw Design Decision Matrix

Hand	Price	Ease	Redundancy	Reliability	Total
Vice	1	2	4	5	12
Double Servo	3	4	3	2	12
Linkage	4	3	3	3	13

Decision: Linkage

3b. Justification for selected components:

The reason behind selecting the ball screw system was primarily due to its simplicity and need for one singular stepper motor to work. Therefore, a stepper motor with enough torque to move the ball screw system was chosen and paired with a compatible driver. Similarly, the claw mechanism was chosen due to the mechanical advantage and simplicity of its design, as it also only required one servo motor to operate efficiently. Furthermore, it was the lightest option of the three, allowing for less resistance to be placed on the ball screw system.

There was not a decision matrix made for the color sensors. However, there is justification as to why these sensors were chosen. The goal of the robot is to move three colored blocks. The TCS3200 color sensor returns three wavelengths in the 470, 550, and 665 nanometer ranges. These correspond with blue, green, and red - three colors for three blocks. This particular sensor was chosen due to its cost. The AS7262 spectral sensor was chosen because it returns six wavelengths: 400, 470, 550, 600, 630, and 665 nanometers. These correspond to violet, blue, green, yellow, orange, and red colors. The colored indicators on the ground consist of five of those colors. The higher cost of the AS7262 is justified due to its increased accuracy.

3c. Trade-offs considered:

The report is expected to discuss trade-offs such as cost, accuracy, robustness, and safety. The team's actual trade-off decisions were seen at several points throughout the project, specifically in regards to the ball screw system. As seen above in *Table (2)* the decision to go with the ball screw required a trade-off in price with the articulated arm and was thought to be more reliable as well. However, the ball screw was still cheaper and easier than the extending arm. After completing the decisions matrix the part with the highest total points is what was chosen. *Table (3)* refers to the decision matrix that was used to determine the hand/claw design. The highest trade-off with choosing the linkage over vice was reliability. Although the vice was decided to be the most reliable out of the three options the linkage was the cheapest. Many other minor trade-offs were made due to budgeting constraints, especially with motor selection. All motors on the robot were sourced from the Reef on campus, as ordering any more was entirely out of the given 100 dollar budget. This meant that the functionality of the robot was sacrificed slightly due to part availability and if given access to a larger budget, less compromises would have been made.

4. Theoretical Analysis and Calculations

4a. Relevant theoretical modeling:

For this project, relevant analysis includes the following: wheel torque and motor sizing, robot speed and travel time, grasping force required to hold a cube, battery/power consumption, basic kinematics of the arm or gripper, sensor threshold logic and calibration relationships.

Wheel Torque and motor sizing were evaluated based on the robot's approximation mass of 2.5 kg and wheel radius of 0.05 m. The force required to move the robot was estimated using rolling resistance and conservative assumptions for acceleration, resulting in an estimated force of approximately 2 N. Using the torque

relationship $T = rF$, the required torque per drive wheel was calculated to be approximately 0.10 N·m. Applying a safety factor of 2-3 to account for inefficiencies and uneven surfaces resulted in a design requirement of approximately 0.2-0.3 N·m, confirming that the selected drive motors are sufficient.

Robot speed and travel time were estimated from motor rotational speed and wheel diameter, allowing linear velocity to be approximated. These estimates ensure that the robot can complete its required tasks within time constraints.

Clamping force was analyzed using a friction-based model to prevent slipping of the cube during lifting. Assuming a worst case cube mass of 0.5 kg, the weight was calculated as 4.9 N. Using a conservative coefficient of friction of 0.3, the required clamping force was determined to be approximately 16.3 N. After applying a safety factor of 2, the design clamping force was increased to approximately 30-35 N. Based on a gripper finger length of 0.05 m, the required servo torque was calculated as approximately 1.63 N·m. This was compared to the available torque of the 20 kg·cm(1.96N·m) servos, confirming that the actuators are capable of providing sufficient gripping force.

The vertical motion of the arm was modeled using a ball screw mechanism. The torque required to lift the load was determined using the relationship:

$$T = \frac{FL}{2\pi\eta} \quad \text{Eqn. 1}$$

Where:

T = Torque (N·m)

F = Load Force (kg calculated to Newtons by multiplied by 9.81 m/s² for gravity)

L = Screw Lead (m, how “long” it takes for a full revolution on ball screw)

η = efficiency

Assuming a load of 1kg, a screw lead of .5 cm/rev, and an efficiency of 0.9, the required torque was calculated to be approximately 0.009N·m. This low value demonstrates the mechanical advantage of the ball screw system, indicating that the selected stepper motor is more than sufficient even when accounting for friction and non-ideal conditions.

Power consumption was also considered, particularly for the servo motor. Using the power relationship:

$$P = T\omega \quad \text{Eqn 2}$$

Where:

P = Power Required (Watts)

T = Torque Required ($N \cdot m$)

ω = Angular Velocity (rad/s)

The peak mechanical power output of the servo was estimated to be approximately 11.6 W based on its torque and angular velocity. Total system power requirements were estimated by summing contributions from motors, sensors, and control electronics to ensure the battery can support both average and peak loads.

Basic kinematics of the arm and gripper were analyzed to ensure the system can reach and manipulate the cube effectively. The ball screw provides controlled vertical motion, while the servo-actuated claw enables grasping. Via Jackson, the sensor threshold logic was developed for calibration of the color and light sensors, allowing the robot to reliably distinguish objects and environmental conditions based on measured signal ranges.

4b. Circuit analysis and wiring diagrams:

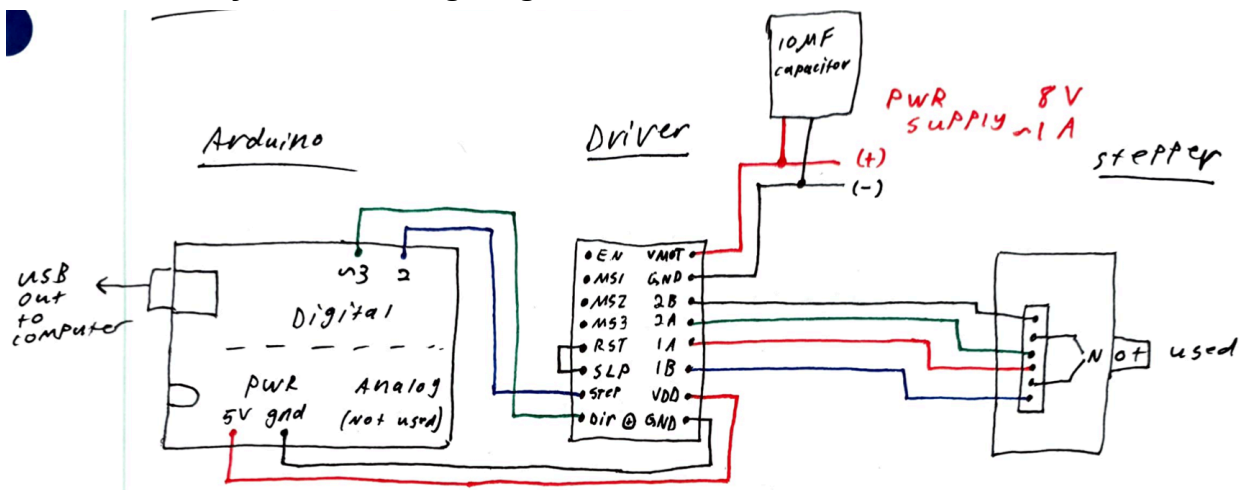


Figure (4): Stepper motor, driver, Arduino, and power source wiring diagram

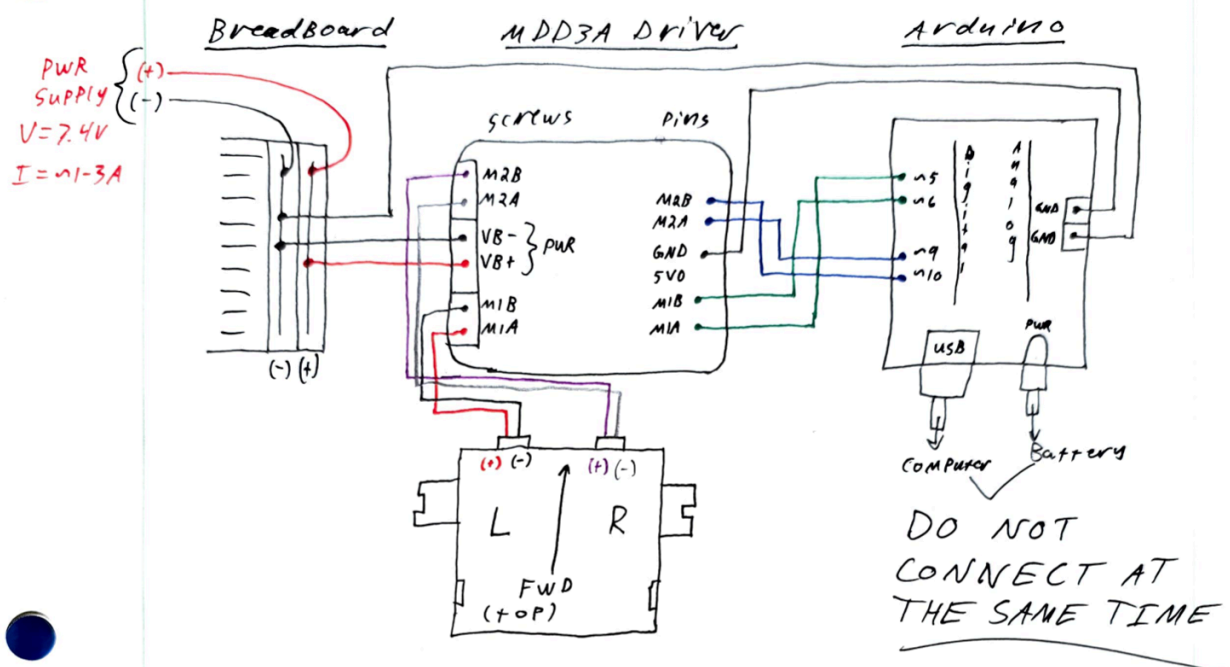


Figure (5): Geared DC wheel motors, driver, Arduino, and power source wiring diagram

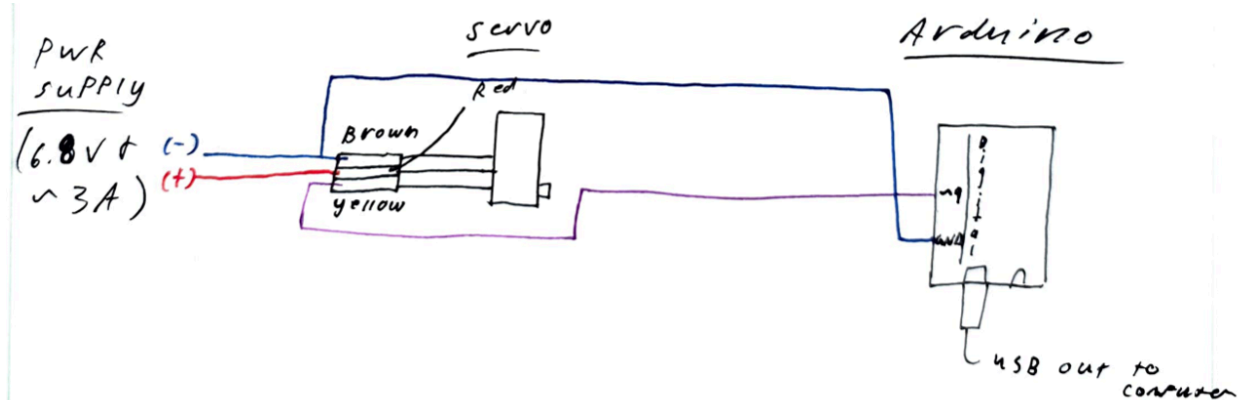


Figure (6): Servo motor, Arduino, and power source wiring diagram

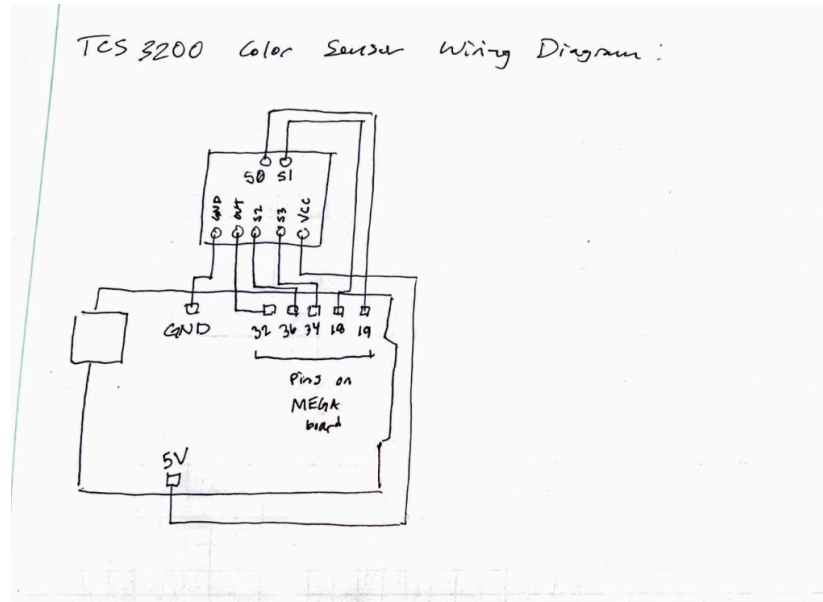


Figure (7): TCS3200 color sensor sensor wiring diagram

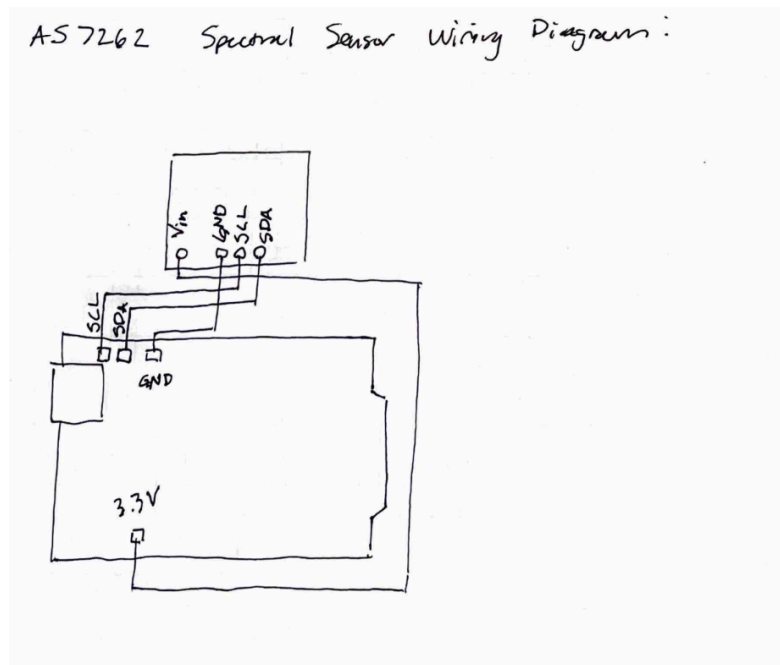


Figure (8): AS7262 spectral sensor wiring diagram

4c. Assumptions and limitations of the analysis:

Several assumptions were made to simplify the analysis. The robot is assumed to operate on a flat, smooth surface, allowing resistance to be treated as constant. Variations in terrain and surface friction were not explicitly modeled. Friction coefficients used in the grasping analysis were estimated based on typical material interactions (for example, PLA on plastic) and may vary in practice. As a result actual gripping

performance may differ depending on the surface conditions and object material. Motor performance was assumed to match manufacturer specifications; however, real world factors such as voltage drops, efficiency losses, and load variations may reduce available torque and speed. Similarly, the ball screw system was assumed to operate with high efficiency and minimal backlash, which may not fully reflect real mechanical behavior.

A key limitation of the system is that it can't rely on cameras or vision based sensing and must instead operate using onboard sensors such as color and light sensors. This limits environmental awareness and makes the system sensitive to ambient lighting conditions, which may affect detection accuracy despite calibration. Battery performance was also assumed to be ideal, although actual capacity may decrease under high loads or prolonged operating conditions. Additionally, dynamic effects such as vibrations, impacts, and mechanical misalignment were not fully accounted for and may influence overall system performance.

5. Implementation and Integration

5a. Mechanical fabrication and assembly:

Based on the design decisions made in the preliminary design phase, we designed the mechanical components to function according to the theoretical analysis and to integrate the electrical components. The model shown below in *Figure (9)* was fabricated using 3D printing with PLA and PETG.

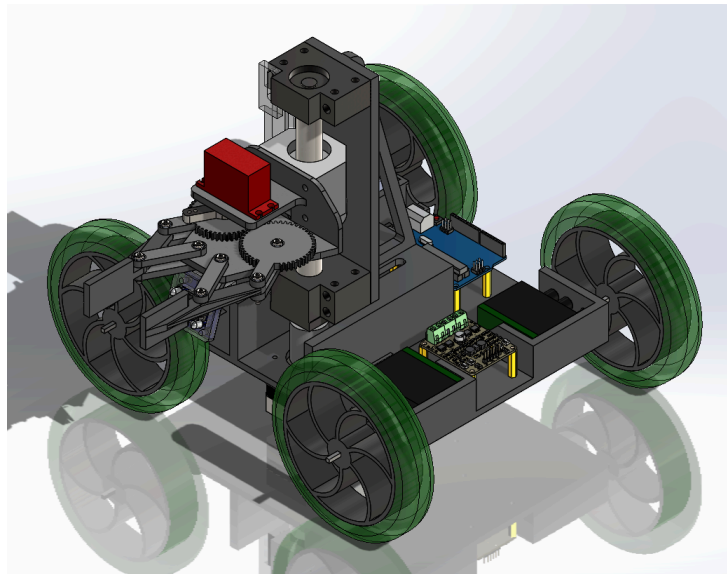


Figure (9): Full System SolidWorks CAD Model

Subsystem: Ball Screw

We decided to buy the ball screw to limit the amount of machine and save money. We had to design a guide, shown in *Figure (11)* below, to ensure it remains straight as the screw spins. This was 3D printed using PETG to ensure the strength will be enough. The rest of the model shown in *Figure (10)* is designed based on the measurements from the purchased component.

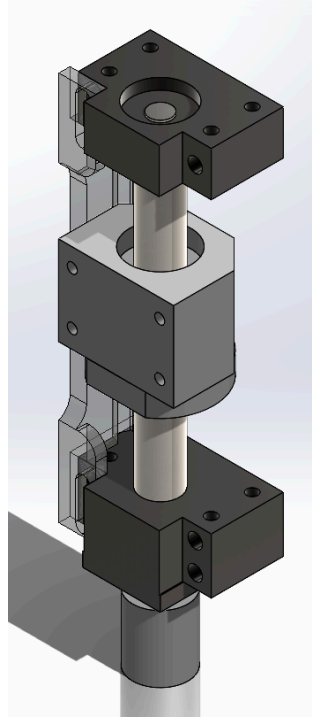


Figure (10): Ball Screw SolidWorks CAD Model

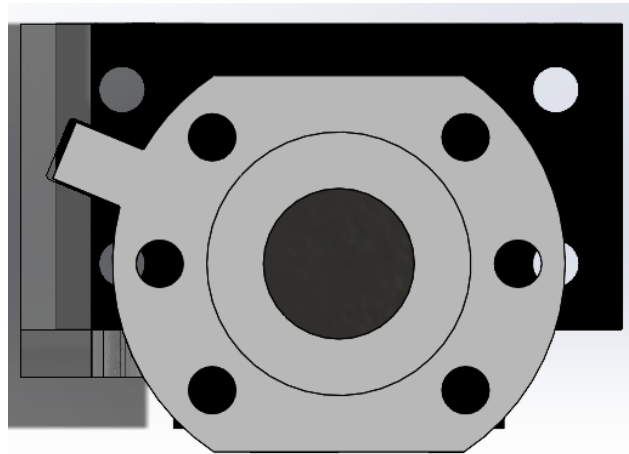


Figure (11): Ball Screw Guide Top Down Section View

Subsystem: Claw

The claw design chosen was a parallel gripper mechanism shown in *Figure (12)* below. This mechanism works by creating a parallelogram with one fixed edge. When

you rotate one of the edges with respect to the fixed edge, it closes the claw while keeping the grips at the end of the claw parallel. The system integrates the 20kg servo motor using a gear to transfer the motion to both arms. Finally, it mounts the color sensor below as seen. All components were fabricated via 3D printing using PLA.

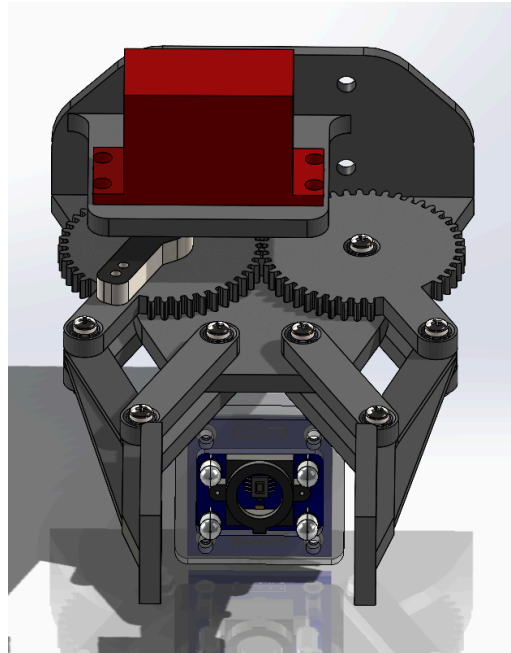


Figure (12): Parallel Gripper Claw Mechanism SolidWorks CAD Model

Subsystem: Chassis

The chassis subsystem shown in *Figure (13)* below was fabricated by 3D printing using PLA. The main purpose of this subsystem was to have a base for all other mechanical subsystems to mount to. It also serves the important purpose of providing a place for the electronic components to mount to, as shown in *Figure (14)*.

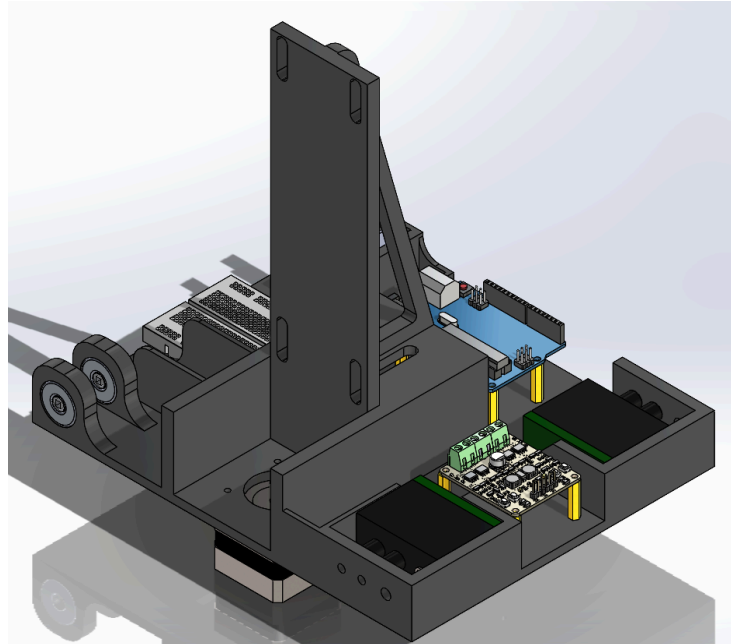


Figure (13): Chassis SolidWorks CAD Model

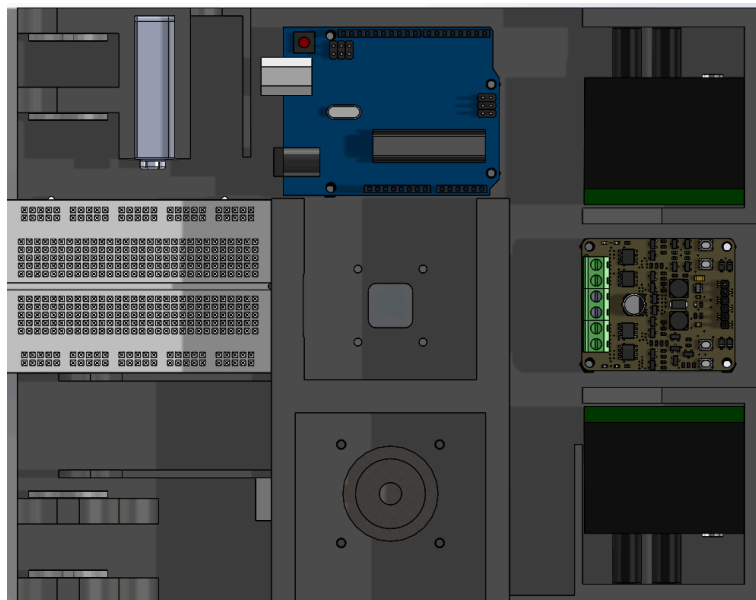


Figure (14): Chassis Electronics Layout

5b. Electrical wiring and power distribution:

The Arduino MEGA is powered by one 9V battery. The Arduino provides power for the stepper driver and claw color sensor, which both share the 5V pin and the 3.3V pin powers the ground color sensor. The wheel motor driver is powered by an independent 7.4V rechargeable battery pack to avoid any power loss from other component draw. Similarly, the stepper motor also has its own independent power source via a 9V battery that was introduced after the initial 7.4V battery failed near the

project's completion. Lastly, the servo motor was independently powered by a quad AA battery pack, which allowed it to operate within its 4.8-6.8V range. More on the actuator power supplies and drivers can be found in section "2d. Actuation subsystem description." All of these systems share a common ground to the Arduino.

5c. Software and control logic implementation:

The control logic is dictated by which colored indicator the line following sensor detects. The robot will start upon detection of yellow and will move forward until it detects orange. This tells the robot that a block needs to be picked up and scanned by the claw color sensor. The robot will move backwards until it hits one of the red, green, or blue colors on the ground. The robot will cross-check whether the held block matches the color on the ground. If true, then the robot will drop the block and increase the count of blocks dropped. The robot will move forward again until it hits orange again, where it will skip over the first orange indicator depending on the number of blocks dropped. (e.g. the robot will pass over two orange indicators if two blocks have already been dropped). This process will repeat until all three blocks have been dropped. While the robot is reversing, it will not react to any orange indicators if the robot holds a block. The program utilizes several Arduino libraries such as servo.h to control the claw servo, and wire.h along with Adafruit AS726x to control the line following sensor.

6. Calibration, Testing, and Validation

6a. Sensor calibration procedure:

Color sensors must be calibrated when there are changes in the environment's lighting. The sensors will return the highest wavelength as a color. There are six wavelengths corresponding to violet, blue, green, yellow, orange, and red. Each wavelength must be adjusted by adding to or subtracting from the raw sensor output so it does not overpower other wavelengths. Each color will be tested until the sensor returns the correct color. For white and black colors, the sum of all sensor output values will be combined to determine the highest and lowest threshold for each.

6b. Test setup and methodology:

The test setup for the robot consisted of calibrating the color sensors if the lighting environment has changed, placing the robot, cups, and target blocks on pre-marked positions, and verifying the robot is aligned with the black line. The robot must be tested on flat ground, and the claw must be at the top position to ensure it does not knock over the target cups or blocks. By ensuring these variables are consistent, the robot should be able to pick up and drop blocks consistently.

Testing for the motors and their respective drivers involved individual test code and a benched power supply to ensure their functionality. Circuit diagrams were drawn up on paper during this test process to regulate the pin allocation for the arduino, drivers, and individual motors. Certain motors, like the stepper for instance, required a very specific power up and power down sequence to avoid damaging its motor driver. This was documented and presented to all group members for their reference. The power up sequence is as follows: complete all wiring to the motor, driver, and Arduino, hook up the motor power supply, plug in the Arduino, and upload code. Similarly, the powering down procedure involved these steps: let the stepper motor stop moving, unplug the Arduino, and disconnect the stepper motor power supply. Doing either of those processes out of order could potentially damage the stepper motor driver. Following that process, the servo was then tested with its designated power supply to verify that it met the stepper motor's voltage and current requirements. Similarly, the servo motor and geared dc motors also underwent this process individually, with all three motor subsystems having circuit diagrams drawn out for future reference. Later on in the project, these subsystems that had been previously tested individually were combined with the others (color sensors and mechanical design aspects) to work as one complete unit. Subsystems were integrated and tested methodically, as only one would be added to the system and tested at a time before adding another.

6c. Discussion of failures and troubleshooting:

There were multiple failures that were encountered during testing. Before the robot was even fully assembled, the claw assembly and ball-screw motor was tampered with and broken. Other large issues we encountered were with the Arduino itself. The library the servo uses to open and close the claw, servo.h library, interfered with PWM pins 9 and 10. The Arduino UNO did not have enough pins to support both this library and the four required PWM pins for the two wheel motors. This required an upgrade to the Arduino MEGA at the last minute. There are also not enough pins providing voltage on the Arduino, so systems like the stepper driver and claw color sensor had to share voltage.

The chassis for the robot also had several issues with several stress concentrations around bolt holes, leading to breakage. There was an uneven distribution of weight on the robot, leading to drifting during driving. The wheel housings weren't completely adequate for the wheels, causing them to drift side to side. The design of the robot also inhibited any line following functions due to having no way to turn. Lastly, the most notable errors encountered were with the electrical systems. The batteries used from the Aquarium were either worn out or dying. The motors have enough power to complete one to two cycles of testing before they begin to slow down -

heavily impeding the repeatability of the test. Along with this, there were issues with wiring being messy and loose. Alongside these major problems were smaller issues that impeded with troubleshooting:

- Ambient lighting interfering with color detection
- Color detection code not reliably picking up certain colors
- General control logic coding errors

There were many compromises made in order to solve these issues. The robot was coded to interact with the black line as a part of its route between the cubes and the cups, however, it has no way to correct itself in the case that it drifts off course. To compensate, the wheel motors were calibrated to move at different speeds - not unlike a car differential. This counteracted the drift from the uneven weight distribution. To solve the electrical issues the robot was plugged in after one or two tests to increase repeatability and the batteries were replaced repeatedly. Replacing the Arduino UNO with the MEGA solved the conflicting PWM pin issues, and distributed more current to the shared 5V rail. Most other errors of note were solved with testing.

6d. Performance metrics and repeatability results:

During testing, the robot was able to complete several full cycles of three blocks consecutively. If calibration is done and all other variables are in check, then the only unreliable aspect of the robot is the speed. As noted previously, the motor batteries are extremely weak, and the speed between two full cycles of testing significantly decreases. This impacts the distance at which the robot moves after detecting a ground color indicator, leading to the robot potentially missing the block or cup entirely. Considering these flaws, the robot can consistently deliver two to three blocks from point A to B.

7. Results and Discussion

7a. Comparison between expected and actual performance:

As outlined in the project description, the robot must pick up three colored cubes at point A and place them in corresponding cups at point B within a specific order. The robot must have a wheeled base, an Arduino controller, and a grasping mechanism. It cannot have any camera or vision systems, and must be powered by an onboard power source.

According to these parameters, our robot meets all of the requirements. The robot was able to autonomously pick up blocks at point A and deliver them to point B without manual correction.

7b. Limitations of the current system:

The main limitation this robot demonstrates is a lack of self-correction. The robot's design inhibits any line-following function, so if the robot begins to drift off the line the robot cannot turn to fix it. This has not posed an issue after testing the robot within three cycles, but the longer the robot runs the chance for error increases. Testing has concluded that the robot can move all three blocks from point A to B in roughly 65 seconds. Altogether, this robot meets all expectations outlined in the project description.

8. AI usage and reflection

8a. AI tools used and purpose of use:

For this project, the primary AI chatbot model used was ChatGPT 5.4 with some code cross-examination from Gemini 3. The primary motives for using these chatbots were threefold. Firstly, ChatGPT 5.4 was used to verify the initial torque and power requirement calculations to ensure their validity before selecting actuators and other components. Secondly, ChatGPT 5.4 also assisted with preliminary compatibility questions regarding motors and their respective drivers, as well as drafting power up and power down sequencing to ensure no damage to the drivers during those procedures. Lastly, these AI tools were also used to work out kinks in the code when the subsystems were combined together. ChatGPT 5.4 and Gemini 3 both analyzed and identified anything that may be an issue.

8b. Engineering questions asked to AI:

AI was used throughout the project as a supplemental engineering resource to assist with troubleshooting, concept evaluation, and technical clarifications. Questions asked to AI primarily focused on areas such as theoretical calculations, sensor and actuator integration, control logic development, component selection considerations, and debugging assistance. AI was also used to help to verify calculation approaches and provide guidance on programming workflows.

8c. Summary of AI suggestions:

AI provided suggestions related to improving system organization, refining control strategies, troubleshooting code, or wiring issues, and considering alternative design approaches. All suggestions generated by AI were reviewed critically by the team and compared against engineering principles, calculations, testing results, and project requirements before implementation. Final design decisions and validation remained a priority to the team.

8d. Critical evaluation of AI limitations or incorrect recommendations:

Limitations of AI were quite prevalent during the group's time utilizing it, specifically in two different areas. One such limitation occurred during the initial stages of selecting components that fit within the budget. ChatGPT 5.4 incorrectly assumed that the group somehow had access to a 1,000 dollar budget for motors and drivers, which led to part sourcing issues early on. The second limitation encountered would be when asked to analyze the complete code for errors, as the chatbot would have a tough time understanding the project parameters. It would begin to make certain incorrect assumptions about the requested prompt. This was observed with the color sensors specifically, as the model would assume that the group only had one color sensor and would incorrectly combine our two separate sensors within its generated code. Similarly, the chatbot would also combine certain motors of different types together in the code, which was less than ideal for the robot's functionality.

8e. How AI influenced design decisions without replacing engineering judgment:

AI influenced design decisions without replacing our engineering judgement by being employed as a supplementary tool to assist with brainstorming, troubleshooting, and improving effectiveness or efficiency during the design process. However, all final engineering decisions were based on engineering principles, calculations, feasibility, and team judgement. AI outputs were evaluated critically rather than accepted directly, ensuring the design remained grounded in engineering analysis and project requirements.

9. Individual Contributions

Owen:

Served as the team lead by keeping the team on track and scheduling things out. Designed, fabricated, and assembled all the mechanical subsystems to ensure that all the subsystems worked together. Ran the testing of the whole system and made the code updates to get the intended functions. Led the solving of problems that arose during the final testing phase.

Cadyn:

Selected appropriate motors that adhered to the given design, budget constraints, and torque requirements provided, researched adequate drivers for said motors, and found power supplies that met the power needs of the robot. Also tested and troubleshooted each individual motor, driver, and power supply before and during the integration with the other subsystems.

Jakson:

Wrote the code that the robot used to detect the block colors along with the code to detect which color the robot is driving over. Implemented the color detection code into the wider robot control logic and assisted with overall troubleshooting.

Sarah:

Wrote original code logic statements and helped to implement the individual subsystems code into the main code loop.

Colton:

Performed theoretical calculations (motor(s) output, clamp/grasping force, and ball screw mechanism) to guide and support design decisions.

10. Ethical, Safety, and Professional Considerations

10a. Safety considerations in design and testing:

The main safety concern considered was the electrical system. Thorough documentation for the order of setting up and taking down the wiring was made. For example, if the arduino is plugged into a power source while simultaneously plugged into a computer it will backfeed voltage into the computer and short it out. Additionally, if certain pins are not removed in a specific order while power is still plugged in, certain components will short out. One other safety concern would be the Arduino power supply voltage and current ratings, as too high of either could potentially start an electrical fire. All of these were taken into account during the design and component selection process to prevent any safety issues from happening.

10b. Responsible and ethical use of AI:

AI tools were used ethically and responsibly, as the group did not have either chatbot listed previously engage in anything improper for the course. Instead, it was used as a learning aid to fill in any gaps in member knowledge during the project durations and assisted in an appropriate fashion when prompted to provide help. More on this can be found in the previous AI usage section above.

11. Conclusion and Future Work

11a. Summary of achievements:

The robot successfully completed the tasks outlined in the project description. To achieve this, our robot can read colored indicators on the ground, drive autonomously between points A and B, detect whether the robot is holding a block and what color that block is, and can determine how many blocks have been picked up.

11b. Lessons learned:

Several lessons were learned while testing the robot. A completely different wheel base is required for the line-following function to be possible. More reliable power sources need to be found that can consistently provide the desired voltage needed to move the robot. Considerations to the chassis design need to be made in order to prevent stress concentrations around areas with heavy torque, as well as better distributing the weight of the parts. Cables were frequently disconnected due to disorganization, meaning better cable management will be needed in the future. Improvements to the efficiency of the code could be made to better increase the responsiveness and reliability of the robot. Overall, mostly every aspect of the robot can be learned from.

11c. Recommendations for future improvements:

As listed in the previous segment there are a few important design improvements to be implemented. These include improving the electrical system, increasing support around areas with heavy load, changing the four wheel design to a design that can oscillate back and forth across a line, and improving the control logic.

12. References

- Miller, Dean. "Adafruit AS7262 6-Channel Visible Light Sensor." *Adafruit Learning System*, learn.adafruit.com/adafruit-as7262-6-channel-visible-light-sensor/arduino-wiring-test. Accessed 1 May 2026.
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